

$$\text{Four } H(t) = 1/p$$

$$t = 1/p^2$$

$$t^n = \frac{n! \rightarrow 1.2.3. \text{ etc}}{p^{n+1}}$$

$$e^{at} = 1/p - a$$

$$\sin(\omega t) = \frac{\omega}{p^2 + \omega^2}$$

$$\cos(\omega t) = \frac{p}{p^2 + \omega^2}$$

$$x(t-a) = e^{-at} \cdot (p)$$

$$\int (t-a) \cdot H(t-a) = e^{-at} - F(p)$$

→ displacement

$$e^{-at} \cdot x(t) = X(p+a)$$

$$1) \lambda'(t) - 2\lambda(t) = 0 \rightarrow \lambda(0) = -1$$

équation différentielle?

PAS
SAVOIR

$$\mathcal{L}(\lambda'(t)) - 2 \mathcal{L}(\lambda(t)) = \mathcal{L}(0)$$

$$\mathcal{L}(\lambda'(t)) = p \cdot \mathcal{L}(\lambda) - \lambda(0)$$

$$\mathcal{L}(\lambda) = S(p)$$

$$p \cdot \mathcal{L}(\lambda) - \lambda(0) - 2 \cdot \mathcal{L}(\lambda) = 0$$

$$S(p) \cdot (p-2) - 1 = 0$$

$$S(p) \cdot (p-2) = 1$$

$$S(p) = \frac{1}{p-2}$$

$$\frac{1}{p-a} = e^{at}$$

$$\lambda(t) = \mathcal{L}^{-1}\left(\frac{1}{p-2}\right) = e^{2t}$$

2)

Dans le cadre ou on a du

1

il faut le diviser en plusieurs partie tel que *

$$p^2(p-2)$$

$$* \frac{A_1}{p} + \frac{A_2}{p^2} + \frac{B_1}{p-2}$$

$$\begin{aligned} \rightarrow (E=0) &\rightarrow \cancel{p^2} \cdot \frac{1}{\cancel{p^2} \cdot (p-2)} \rightarrow \frac{1}{p-2} \rightarrow \frac{1}{0-2} \rightarrow -\frac{1}{2} \\ &\quad \downarrow \\ &\quad A_2 = -1/2 \end{aligned}$$

$$\begin{aligned} \rightarrow (E=1) &\rightarrow \cancel{p-2} \cdot \frac{1}{p^2(p-2)} \rightarrow \frac{1}{p^2} \rightarrow \frac{1}{4} \rightarrow B_1 = 1/4 \\ &\quad \downarrow \\ &\quad p=2 \end{aligned}$$

quand
 $\frac{A_1}{p}$

$$\rightarrow A_1 \rightarrow 1 = A_1 - A_2 + B \rightarrow A_1 = 1 - (-1/2) + 1/4 = -1/4$$

$$= -\frac{1}{4p} - \frac{1}{2p^2} + \frac{1}{4(p-2)}$$

normal

esce

$$b) (t^2 + 1)^2 = (t^4 + 2t^2 + 1) = \frac{24}{t^5} + \frac{4}{t^3} + \frac{1}{t}$$

$r > a^2 + 2ab - b^2$

$$g) H(t-1) = e^{-t} \cdot \mathcal{L}(H(t)) = e^{-t} \cdot \frac{1}{t} = \frac{e^{-t}}{t}$$

$r > 1/t$

$\mathcal{L}(t-a) = e^{-at} \cdot (P)$

$$R) (t-1) \cdot (H(t-1)) = e^{-t} \cdot \frac{1}{t^2} = \frac{e^{-t}}{t^2}$$

$\mathcal{L}(t-a) \cdot H(t-a) = e^{-at} \cdot F(P)$

$$h') (t-3) \cdot (H(t-3)) = \frac{e^{-3t}}{t^2}$$

$$j) (t-5) \cdot (H(t-4)) = (t-4) - 1 \cdot (H(t-4)) = e^{-4t} \cdot \frac{1}{t^2} - \frac{1}{t}$$

$$k) e^{3t} \cdot t^3 = \frac{6}{(t-3)^4}$$

$r > 6/t^4$

$e^{-at} \cdot \mathcal{L}(t) = X(t+a) \quad (t-3)^4$

$a = -3$

$$l) \quad e^t \cos(t) = \frac{1}{\omega^2 + p^2} = \frac{(1-1)}{\omega^2 + (1-1)^2} = \frac{1-1}{(1-1)^2 + 1}$$

$$m) \quad e^{-4t} \cdot \sin(5t) = \frac{5}{(4+4)^2 + 25}$$

$$n) \quad e^{-2t} (t^2 + t + 1) = \frac{2}{(p+2)^2} + \frac{1}{(p+2)^2} + \frac{1}{p+2}$$

$$o) \quad e^{-2t} (t^2 - 1)^2 = t^4 - 2t^2 + 1 \cdot e^{-2t} = \frac{24}{(p+2)^5} + \frac{1}{p+2} - \frac{4}{(p+2)^3}$$

$$p) \quad 2e^{-5t} \cdot (\cos(2t) + \sin(2t)) = 2 \cdot \frac{(p+5) + 2}{(p+5)^2 + 4} = \frac{2p + 14}{(p+5)^2 + 4}$$

inverse

$$a) \frac{3}{s+2} - \frac{1}{s^3} = 3e^{-2t} - \frac{t^2}{2}$$

$$\rightarrow \frac{s+2}{s+3} - \frac{-4+2}{-4+3} = 2$$

$$b) \frac{s+2}{(s+3)(s+4)} = \frac{A_1}{s+3} + \frac{A_2}{s+4} = \mathcal{L}^{-1}\left(\frac{1}{s+3}\right) + \mathcal{L}^{-1}\left(\frac{1}{s+4}\right) = -e^{-3t} + 2e^{-4t}$$

$$CE = -3 \rightarrow \cancel{s+3} \cdot \frac{s+2}{\cancel{s+3} \cdot (s+4)} = \frac{s+2}{s+4} = \frac{-3+2}{-3+4} = -1 = A_1$$

$$c) \frac{a}{s^2 - a^2} = \frac{a}{(s+a)(s-a)} = \frac{A_1}{s+a} + \frac{A_2}{s-a} = \frac{-1}{2} e^{-at} + \frac{1}{2} e^{at}$$

$\downarrow CE = -a$ $\downarrow CE = a$
 $\frac{a}{s+a} = \frac{a}{-2 \cdot a} = -\frac{1}{2}$ $\frac{a}{s-a} = \frac{a}{2 \cdot a} = \frac{1}{2}$

$$d) \frac{3}{(s+6)^2} = 3e^{-6t} \cdot t$$

$$\rightarrow -1 \quad \rightarrow 1$$

$$e) \frac{10}{s^2 + 4s - 24} = \frac{10}{(s+7)(s-3)} = \frac{A_1}{s+7} + \frac{A_2}{s-3} = \frac{-1}{s+7} + \frac{1}{s-3} = -e^{-7t} + e^{3t}$$

$\Delta = 5^2$
 $2 \cdot 3 \rightarrow \Delta \cdot (s - x_1) \cdot (s - x_2)$

$$f) \frac{1}{(s-5)^3} = \frac{(s-5) + 5}{(s-5)^3} = \frac{s-5}{(s-5)^3} + \frac{5}{(s-5)^3} = \frac{1}{(s-5)^2} + \frac{5}{(s-5)^3} = e^{5t} \cdot t + e^{5t} \cdot \frac{5t^2}{2}$$

$$g) \frac{s+7}{s^2+4s+9} = \frac{1}{s^2+4s+9} + \frac{7}{s^2+4s+9} = \cos(7t) + \sin(7t)$$

$$h) \frac{e^{-s}}{s^3} = \frac{(t-1)^2}{2}$$

$$a=1 \quad e^{-s} \cdot \frac{1}{s^3}$$

$$-X(t-a) = e^{-at} \cdot X(p) \quad \frac{t^2}{2}$$